

```
# 1. read csv file
import pandas as pd
df = pd.read_csv("wuzzuf_jobs.csv")
# can add ( ,header = none)
# ( , skiprow = 1)
# ( , skipfooter = 3)
```

```
# 2. read top 10 rows
df.head(10)
```

	Unnamed: 0	TimeStamp	Title \
0	0	2024-02-19 20:09:36.277000	Dotnet Developer
1	1	2024-02-19 20:10:43.450000	Android deve
2	2	2024-02-19 20:23:24.387000	Android Developer
3	3	2024-02-19 20:27:08.358000	DevOps Engineer
4	4	2024-02-19 20:31:36.432000	Java
5	5	2024-02-19 20:32:35.324000	R&D computer vision
6	9	2024-02-19 20:49:55.889000	Dot net developer
7	10	2024-02-19 20:59:43.384000	raact developer
8	11	2024-02-19 21:02:18.888000	Full Stack developer
9	13	2024-02-19 21:04:02.696000	Data Analysis

	Years of Experiences	Salary	Date of Salary	What Is your
Company \				
0	2	8000	2024-02-01 00:00:00	
Egyption				
1	NaN	6000	2024-02-06 00:00:00	
Egyption				
2	1	10000	2024-02-20 00:00:00	
Egyption				
3	1 month	10000	2024-02-19 00:00:00	
Egyption				
4	1	20000	2024-02-19 00:00:00	
Egyption				
5	1	15000	2023-07-16 00:00:00	
Egyption				
6	2 months	6000	2024-02-29 00:00:00	
Egyption				
7	1	15000	2024-03-01 00:00:00	
Egyption				
8	2	5000	2024-02-28 00:00:00	
Egyption				
9	1	6000	2024-01-01 00:00:00	
Egyption				

	Work Type	Work Hour	City of	Company site
0	Hybird	Full Time		Cairo
1	Hybird	Full Time		Assiut
2	Hybird	Full Time		Cairo
3	On Site	Full Time		Assiut

4	On Site	Full Time	Cairo
5	Remotley	Full Time	Cairo
6	On Site	Full Time	Mansoura
7	Remotley	Full Time	Cairo
8	Remotley	Full Time	Cairo
9	On Site	Full Time	Banha

```
# 3. print the last 5 rows
df.tail()
```

Unnamed: 0		TimeStamp	
Title \			
903	1323	2024-03-07 10:04:23.532000	Software developer
904	1324	2024-03-07 12:25:44.049000	Embedded SW Engineer
905	1325	2024-03-07 12:52:53.200000	Php developer
906	1329	2024-03-08 19:01:07.101000	Seniors interior design
907	1330	10:10:49 2024/03/02 ص	DevOps Engineer

Years of Experiences	Salary	Date of Salary	What Is your Company \
4	45K	2024-02-01 00:00:00	903
0	10k	2024-03-01 00:00:00	Egyption
1	10000	2024-03-01 00:00:00	904
7	19000	2024-01-01 00:00:00	Egyption
1	29000	2024-03-03 00:00:00	905
			Egyption
			906
			Egyption
			907
			Egyption

Work Type	Work Hour	City of Company	site
Remotley	Full Time	Cairo	903
On Site	Full Time	cairo	904
Remotley	Full Time	Cairo	905
Remotley	Full Time	Cairo	906
Hybird	Full Time	cairo	907

```
# 4. print information about dataset
df.info()
```

```
# cursor.close()
# connection.close()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 908 entries, 0 to 907
```

```
Data columns (total 10 columns):
#      Column      Non-Null Count  Dtype
---  -
0      Unnamed: 0      908 non-null    int64
1      TimeStamp        908 non-null    object
2      Title             908 non-null    object
3      Years of Experiences  907 non-null    object
4      Salary            908 non-null    object
5      Date of Salary     908 non-null    object
6      What Is your Company  908 non-null    object
7      Work Type         908 non-null    object
8      Work Hour         908 non-null    object
9      City of Company site  816 non-null    object
```

```
dtypes: int64(1), object(9)
```

```
memory usage: 71.1+ KB
```

5. Integration

1. concatenation

```
df_1 = df[['Title' , 'Salary' , 'Work Hour']].sample(5)
```

```
df_2 = df[['Title' , 'Salary' , 'Work Hour']].sample(5)
```

```
df_3 = df[['Title' , 'Salary' , 'Work Hour']].sample(5)
```

```
df_1, df_2,df_3
```

```
(
      Title      Salary  Work Hour
104      front end      8000  Full Time
734  Fullstack Developer (laravel & flutter)  11000  Full Time
858      Unity Game Developer      0  Full Time
673      Software Engineer    15000  Full Time
232      Solution Architect  250000  Part Time,
      Title Salary  Work
Hour
528  Quality Control chemist at pharmaceutical comp...  5000  Full
Time
548      Teacher Assistant      8000  Full
Time
211      ML Eng      8000  Full
Time
835      Full Stack Web Developer    7000  Full
Time
236      Interior designer    3500  Full
Time,
      Title Salary  Work Hour
631  Software developer    7000  Full Time
792  Backend Engineer     30k  Full Time
643  Java Developer    12250  Full Time
307  Biotechnology    10000  Part Time
851  .Net developer     32k  Full Time)
```

```
# concatenation by column
```

```
df_conct = pd.concat([df_1 , df_2 , df_3] , axis = 0)
df_conct
```

		Title	Salary	Work
Hour				
104		front end	8000	Full
Time				
734	Fullstack Developer (laravel & flutter)		11000	Full
Time				
858	Unity Game Developer		0	Full
Time				
673	Software Engineer		15000	Full
Time				
232	Solution Architect		250000	Part
Time				
528	Quality Control chemist at pharmaceutical comp...		5000	Full
Time				
548	Teacher Assistant		8000	Full
Time				
211	Ml Eng		8000	Full
Time				
835	Full Stack Web Developer		7000	Full
Time				
236	Interior designer		3500	Full
Time				
631	Software developer		7000	Full
Time				
792	Backend Engineer		30k	Full
Time				
643	Java Developer		12250	Full
Time				
307	Biotechnology		10000	Part
Time				
851	.Net developer		32k	Full
Time				

```
# concatenation bu row
```

```
df_conct2 = pd.concat([df_1 , df_2 , df_3] , axis = 1)
df_conct2
```

		Title	Salary	Work	Hour	\
104		front end	8000	Full	Time	
734	Fullstack Developer (laravel & flutter)		11000	Full	Time	
858	Unity Game Developer		0	Full	Time	
673	Software Engineer		15000	Full	Time	
232	Solution Architect		250000	Part	Time	
528		NaN	NaN		NaN	
548		NaN	NaN		NaN	
211		NaN	NaN		NaN	

835	NaN	NaN	NaN
236	NaN	NaN	NaN
631	NaN	NaN	NaN
792	NaN	NaN	NaN
643	NaN	NaN	NaN
307	NaN	NaN	NaN
851	NaN	NaN	NaN

		Title	Salary	Work
Hour \				
104		NaN	NaN	
NaN				
734		NaN	NaN	
NaN				
858		NaN	NaN	
NaN				
673		NaN	NaN	
NaN				
232		NaN	NaN	
NaN				
528	Quality Control chemist at pharmaceutical comp...		5000	Full
Time				
548	Teacher Assistant		8000	Full
Time				
211	Ml Eng		8000	Full
Time				
835	Full Stack Web Developer		7000	Full
Time				
236	Interior designer		3500	Full
Time				
631		NaN	NaN	
NaN				
792		NaN	NaN	
NaN				
643		NaN	NaN	
NaN				
307		NaN	NaN	
NaN				
851		NaN	NaN	
NaN				

	Title	Salary	Work	Hour
104	NaN	NaN		NaN
734	NaN	NaN		NaN
858	NaN	NaN		NaN
673	NaN	NaN		NaN
232	NaN	NaN		NaN
528	NaN	NaN		NaN
548	NaN	NaN		NaN

211		NaN	NaN	NaN
835		NaN	NaN	NaN
236		NaN	NaN	NaN
631	Software developer	7000	Full Time	
792	Backend Engineer	30k	Full Time	
643	Java Developer	12250	Full Time	
307	Biotechnology	10000	Part Time	
851	.Net developer	32k	Full Time	

2. Merging

```
df_1 = df[['Title' , 'Salary']][0:5]
df_1
```

	Title	Salary
0	Dotnet Developer	8000
1	Android deve	6000
2	Android Developer	10000
3	DevOps Engineer	10000
4	Java	20000

```
df_2 = df[['Title' , 'Date of Salary']][0:5]
df_2
```

	Title	Date of Salary
0	Dotnet Developer	2024-02-01 00:00:00
1	Android deve	2024-02-06 00:00:00
2	Android Developer	2024-02-20 00:00:00
3	DevOps Engineer	2024-02-19 00:00:00
4	Java	2024-02-19 00:00:00

merge

```
df_merge = pd.merge(df_1 , df_2, on = ['Title'] , how = 'inner')
df_merge
```

```
# inner      shared only
# left       all rows of Table 1
# right      all rows of Table 2
# outer      all rows of both
```

	Title	Salary	Date of Salary
0	Dotnet Developer	8000	2024-02-01 00:00:00
1	Android deve	6000	2024-02-06 00:00:00
2	Android Developer	10000	2024-02-20 00:00:00
3	DevOps Engineer	10000	2024-02-19 00:00:00
4	Java	20000	2024-02-19 00:00:00

3. join

```
df_1 = df[['Title']][0:5]
df_1
```

	Title
0	Dotnet Developer
1	Android deve
2	Android Developer
3	DevOps Engineer
4	Java

```
df_2 = df[['Salary']][0:5]
df_2
```

	Salary
0	8000
1	6000
2	10000
3	10000
4	20000

```
# join
df_1.join(df_2 , how = 'inner')
```

	Title	Salary
0	Dotnet Developer	8000
1	Android deve	6000
2	Android Developer	10000
3	DevOps Engineer	10000
4	Java	20000

```
df_1.join(df_2 , how = 'outer')
# how = 'left'
# how = 'right'
```

	Title	Salary
0	Dotnet Developer	8000
1	Android deve	6000
2	Android Developer	10000
3	DevOps Engineer	10000
4	Java	20000

```
# 4. Data Statistics
df.describe()
```

	Unnamed: 0
count	908.000000
mean	669.802863
std	380.612220
min	0.000000
25%	341.750000
50%	671.500000
75%	993.250000
max	1330.000000

5. Non-Numerical Attributes (text/boolean)

```
df['Title'].value_counts()
```

```
Title
Data Analyst                                10
Frontend developer                          8
Software Engineer                           8
Flutter developer                           7
.net developer                              7
..
Frontend Developer | React Developer         1
Data analyst and operation executive          1
Fluuter developer                            1
Product specialist                           1
Quality control and automation engineer       1
Name: count, Length: 704, dtype: int64
```

6. → What are the different elements?

```
df['Title'].unique()
```

```
array(['Dotnet Developer ', 'Android deve', 'Android Developer',
      'DevOps Engineer ', 'Java ', 'R&D computer vision ',
      'Dot net developer', 'raact developer ', 'Full Stack developer',
      'Data Analysis', 'BI developer ', 'Full Stack developer',
      'Backend .netasp', 'Data Engineer', '1', 'Android developer ',
      'php', 'Data Analyst', 'iOS developer', 'Dot net developer ',
      'Data scientist ', 'IoT embedded SW engineer',
      'Sr. Front End Developer ', 'PHP Developer',
      'computer science instructor', 'Full Stack .NET | Angular |
MSSQL',
      'ML instructor for kids ', '.Net Developer', 'Backend
developer',
      'Android', 'Flutter developer', 'Senior backend .netdeveloper',
      'Laravel developer ', 'Oracle developer ', 'Java Developer',
      'Angular Developer ', 'Junior web developer',
      'Front end developer ', 'MERN Stack developer',
      'Front end developer angular ', '.net full stack ',
      'Back end developer (Node.js) ', 'لاخ', 'React developer ',
      'Data Engineer ', ' Software Engineer ',
      'Senior Full Stack developer ', 'Software Engineering Lead ',
      'Software testing', 'Oracle DBA ', 'Frontend Developer ',
      'Data Integration', 'Data analyst ', 'data scientist',
      'Full stack (Java & React)', 'Erp developer', 'Flutter
Developer ',
      'Data Analyst ', 'Frontend developer', 'Software Developer',
      'BI Developer', 'Machine Learning Engineer ',
      'Software engineer (Backend Java)', '.Net full stack ',
```


'Dot Net Developer ', '.net developer', 'php developer ',
 'Software Developer ', 'DevOps ', '.net developer ',
 'data engineer', 'junior software tester', 'Ios developer ',
 'IOS Developer', 'Software tester', 'Android Developer ',
 'Software Tester', 'Data Analytics engineer',
 '. Net Fullstack developer ', 'BI developer', 'Flutter
 developer ',
 'Business Development Engineer', 'Java developer ',
 'database adminstration', 'Computer vision', 'Frontend React',
 'frontend developer react', 'java spring dev',
 'Netsuite Technical Developer (Javascript Developer)',
 'front end ', 'Software engineer ', 'Machine learning
 engineer',
 'Flutter Developer', 'Lead Mobile Engineer ', ' NodeJS
 Developer',
 'Fullstack Developer ', 'Frontend developer (angular)',
 'Data engineer ', 'data analysis', 'Frontend vuejs',
 '.net full stack developer', 'Vuejs Developer ',
 'Software engineer nodejs, react native',
 'Software engineer stack java ', 'React Developer',
 'PHP Developer ', 'Php developer ', '.NET developer backend',
 'Dotnet developer ', 'Php developer laravel', 'DevOps',
 'Software Engineer', 'Full Stack Python and React',
 'NodeJs Developer', 'Django ', 'React Developer ',
 'Nodejs Backend Developer', 'React native developer',
 'data engineer ', 'Junior Oracle DBA', 'Doctor',
 'Full Stack Developer', 'Social media specialist',
 'Textile Graphic Designer', 'Call Center Representative',
 'Procurement specialist ', 'Test', 'Social media moderator ',
 'Operations manager ', 'Working-drawing architect ',
 'nopde js developer ', 'Java spring Developer',
 'Software Testing Engineer ', 'Sales, telemarketer',
 'Graphic designer ', 'Python developer', 'Petroleum',
 '.NET Developer ', 'Data analyst', 'Tourism sales',
 'Backend SW Engineer', '...', 'Account manager ', 'HR personnel
 ',
 'Customer Service Representative ', 'Merchandising planner',
 'Sales engineer', 'PHP/wordpress developer', 'CRM Analyst',
 'Junior Graphic Designer ', 'Sales executive ',
 'Network engineer ', 'محاسب عام', 'Civil Engineer ', 'UI UX',
 'Branding Designer', 'Senior associate', 'Full stack
 developer',
 'Backend laravel ', 'Senior Social Media editor ',
 'T24 Consultant', 'SEO Specialist ', 'Btl marketing',
 'Junior architect ', 'Subsea inspection engineer ',
 'Civil technical office engineer ', 'Banker ', 'Chef
 accountant',
 'Graphic designer', 'HR HED', 'Nursing ',
 'Software Quality Engineer ', 'General Accountant',

'Outsystems developer', 'Java Deveoper', 'Senior sales engineer',
 ',
 'Full stack .net developer ', 'Full Stack .NET Developer',
 'Content creator/social media specialist ', 'Aws engineer ',
 'Medical repersentive ', 'Senior software engineer',
 'Junior graphic designer ', 'ML Eng ', 'AI engineer',
 'Data analysis ', 'Operation engineer ',
 'Social media specialist ', 'Call centre Germany ',
 'Deputy section head in sales', 'Call center ', 'Representative',
 ',
 'Sales', 'Technical support ', 'Civil engineer', 'Pharmacist ',
 'Frappe Developer', 'Administrator ', 'Faculty of commerce ',
 'Ai Engineer ', 'Seals', 'Senior full stake developer ',
 'Sustainability and ESG Specialist', 'Broker ',
 'Solution Architect ', 'Front-End Developer', 'Senior architect',
 ',
 'Full-stack developer', 'Interior designer ',
 'Machine Learning Engineer', 'Bachelor's degree ', 'Medical
 rep',
 'Agricultural engineer', 'Odoo Developer', 'Marketing
 executive',
 'Front end developer', '.Net software developer ',
 'Electrical engineer ', 'frontend developer', 'Network Engineer',
 ',
 'System engineer ', 'Senior Node js developer', 'Senior
 frontend ',
 'Physiotherapist', 'Cad Designer', 'IT compliance specialist ',
 'Technical office engineer', 'English Instructor', 'N ',
 'Chemist',
 'Android developer', 'Accounting manager ', 'DevOps Engineer',
 'Backend Developer - Django ', 'Media buyer',
 'back-end developer ', 'Procurement specialist',
 'Software Test Engineer', 'Accounting manager',
 'Talent acquisition specialist ', 'ELECTRICAL ENGINEER ',
 'Mobile developer ', 'RPA Developer ', 'Marketing manager ',
 'Front end ', 'Radiologist', 'Logistics assistant ',
 'Software engineer (pyside)', 'I', 'Frontend engineer ',
 'IT specialist ', 'Pharmacist', 'NLP engineer ',
 'Full-Stack (.Net, React)', 'Production Engineer ',
 'Physiotherapy ', 'Cost control engineer', 'Writer ',
 'Senior planning engineer ', 'Engineer ', 'Wep developer ',
 'Database Administrator ', 'Electrical engineer',
 'Biotechnology ',
 'Senior Contracts Engineer', 'Junior Data Scientist ', 'Php',
 'Product Designer ', 'Business Intelligence Analyst',
 'System Admin', 'Medical Representative',
 'Online English Instructor', 'Ai developer ', 'Dentist',
 'Quality Control', 'Senior GIS ', 'Healthcare quality ',
 'iOS Developer', 'Veterinarian ', 'Community pharmacist ',

'Hr manager', 'Backend Node.js', 'h', 'Sales real estate ',
 'Mechanical engineering ', 'Creative Director ', 'Ui/ux
 designer',
 'Planning Engineer (Controls)',
 'Business analyst and Bi developer', 'Big Data Engineer',
 'Credit risk Team leader', 'Juinor front-end developer ',
 'SALES',
 'Php developer', 'iOS Engineer ', 'Next.js Developer',
 'Customer success account manager ', 'Physical therapist ',
 'Restaurants maneger ', 'Django Software Engineer',
 'brand developer', 'accountant ', 'Project control engineer',
 'Senior Software Engineer', 'Lawyer ', 'Junior Frontend
 Engineer ',
 'Brogrammers', 'Fullstack dotnet developer', 'Doctor ',
 'SS Account Manager ', 'Head of Products', 'Pharmacist, gov',
 'Senior travel consultant ', 'Accountant', 'Ai developer',
 'Agriculture engineer', 'IT support', '.NET Developer', 'Desk
 top',
 'Engineer irrigation ', 'Planning engineer', 'Marketing ',
 'Data management engineer', 'Deputy branch head',
 'IT System Administrator ', 'Reliability engineer',
 'Php full stack', 'Sr.full stack developer php', 'Dc ',
 'Otolaryngology consutant (MD degree)', 'Java Developer ',
 'Project manager (site engineer)', 'Junior odoo dev', 'COO',
 'Systems Engineer ', 'Senior Product Designer',
 'Backend Engineer ', 'Java developer', 'Software developers',
 'Dental Technologist ', 'Senior Site Reliability Engineer',
 '..',
 'Accounting ', 'Csr ', 'Customer Service Executive ',
 'Senior android developer ', 'Backend developer ',
 'Mid level ios software engineer ', 'Senior Key Account Manager
 ',
 'Oracle Consultant ', 'Backend java developer',
 'Property consultant ', '.NET intern', 'Performance Analyst',
 '.Net developer', 'IT business analyst ', 'Assistant of
 pharmacy ',
 'Senior consultant media monitoring ', 'Senior iOS developer ',
 'Python developer ', 'Automation Engineer ',
 'Laravel Full stack Developer', 'Administration ', 'Ui Ux ',
 'Junior business analyst ', 'sr Frontend developer',
 'Mfesh mn mfesh', 'technical office engineer',
 'Collection officer ', 'Hr special ', 'nnn',
 'Quality control manger ',
 'Junior Fullstack Developer Laravel Vue', 'Data Scientist',
 'Cybersecurity Engineer',
 'B2b2c customer portal senior specialist ',
 'Agriculture engineering ', 'Coding Writer (Data Processing)',
 'Data Analytics Engineer ', 'Full stack web developer',
 'embedded software engineer ', 'Senior quality control',

'Full stack Engineer ', 'Full stack software engineer ',
 'Senior Java Developer ', 'Penetration tester ',
 'Nodejs developer', 'Senior Project Manager',
 'Oracle apex developer ', 'KG teacher', 'Teacher Assistant',
 'Digital marketing ', 'English teacher', '. Net Developer ',
 '.n',
 'Accounting Manager', 'BI engineer ', 'Coding instructor',
 'Frontend Developer', 'Civil Procurement Engineer',
 'Medical representative ',
 'Senior mechanical quality control engineer ',
 'Senior accountant ', 'php pt y js laravel ajax sql db html css
 ',
 'Technical director ', 'Front end intern ', 'Hr clerk ',
 'WordPress developer', 'Developer', 'junior frontend
 developer',
 'laptop maintenance', 'Physical Security Manager',
 'Devops & linux', 'Oracle HCM Functional Consultant ',
 'performance analyst', 'Marketing executive ',
 'IT Application Support Senior Specialist',
 'Senior Application Support Engineer', 'Software resting level
 2',
 'Seo specialist ', 'Asp.Net Backend Developer', 'Gis and
 survey',
 'Web', 'Systems engineer ', 'Architect',
 'Teaching Assistant architecture ', 'front end',
 'Medical analysis specialist', 'C++', 'Homeroom teacher ',
 'Senior java software engineer ', 'Application support engineer
 ',
 'Quality Control chemist at pharmaceutical company ',
 'Devops Engineer', 'Quality Assurance Engineer', 'Engineer',
 'Application support lead ', 'Site Engineer ',
 'Senior Android developer ', 'Site manager ', 'Lead',
 'Teaching assistant', 'Junior Application support officer',
 'Community Pharmacist ',
 'Business Applications Support Team Leader ', 'Senior
 copywriter',
 'Business applications support engineer ',
 'HRMS Senior specialist', 'Wealth manager cib',
 'Full-Stack Engineer', 'Business analyst',
 'Junior Python Developer', 'QC QA mechanical engineer ',
 'Frontend (React developer)', 'Data Science & Analytics',
 'Blockchain Developer (Web3)', 'Bancassurance ',
 'Junior Application Support ', 'Data Scientist ',
 'Application Support specialist ', 'General Accountan ',
 'Linux System administrator ', 'Php backend developer ',
 'Application Support Engineer ', 'Accountant ',
 'Technical Support Engineer', 'Senior Outsyste ms developer ',
 'Senior full stack java',
 'English teacher at international school ', 'Mechanical

engineer',
 'MERN stack developer', 'Data Management Engineer ',
 'Senior Quality Architect ', 'Senior Flutter Developer',
 'Full stack .net developer', 'Full-Stack Developer',
 'MERN full stack developer ', 'Art instructor ',
 'Oracle fusion administrator ',
 'Back End Developer (Ruby on Rails)', 'Jr java developer ',
 'Technical support engineer (software)',
 'Embedded Software Engineer', 'Senior system engineer ',
 'front end developer', 'Logistics Specialist ',
 'Java Full stack Developer', 'Food',
 'Software Quality Assurance Engineer', 'community pharmacist ',
 'Sr. BI developer', 'Nurse', 'It super visor',
 'Full stack developer (MERN) + React Native ',
 'oracle consultant ', 'civil engineer', 'Full stack developer
 ',
 'Geologist ', 'React Software Engineer',
 'Sales & social media Marketing specialist ',
 'Application support specialist ', 'Software Engineer ',
 'Approval manager', 'Data modeler ', 'Embedded developer',
 'System integration developer ', 'Mechanical site engineer ',
 'JavaScript Developer ', 'Oracle developer', 'Call center agent
 ',
 'Full-Stack Engineer(Spring, React)',
 'real estate graphic designer ', 'Logistics Coordinator ',
 'Qc chemist', 'Frontend angular', 'Software developer',
 'Machine learning Engineer', 'Dermatologist ',
 'Application Support ', 'English instructor ',
 'projects engineer ', 'Clinical information solutions
 specialist',
 'Junior Asp.net corr developer ', 'Node.js Backend Developer',
 'Dot Net Developer', 'Operation Analyst',
 'Application Support linux and application functionality ',
 'Front-end Engineer ', 'Team Lead', 'Full stack .NET
 developer',
 'Sales man in an electrical appliances store ',
 'Senior Application support', 'Full Stack Developer ',
 'Odoo Implementer', 'Senior Data Analyst',
 'Customer service manager ', 'graphic designer ',
 'Engineering manager ', 'Frontend Engineer ',
 'Cyber security engineer ', 'Integration Specialist ',
 'PHP&mySQL',
 'Sales specialist ', 'Devops ', 'C++ developer ',
 'Coding Instructor', '.Net developer ', 'Architect ',
 'Full Stack Web Developer ', 'اخصائي مساحة',
 'Application Support Engineer', 'Quality assurance
 especialist',
 'Senior application support ', 'Copy writer ',
 'Senior Graphic Designer ', 'Sr. RPA Developer',

'Backend Developer', 'Sr. Technical Support (NOC Specialist)',
 'Oil and gas construction engineer', 'طالب بالقوات المسلحة',
 'Site civil engineer', 'Ux ui designer', 'Pr',
 'PHP Back End Developer', 'Software test engineer ',
 'Sr. applications support ', 'Business Intelligence Engineer',
 'Drilling engineer', 'Hr business partner ',
 'مدينة العبور_ القاهرة', 'HR Section Head',
 'Software developer - spring boot , angular',
 'Cable Testing engineer ', 'Network engineer',
 'Senior motion graphic designer', 'Frontend developer ReactJS',
 ',
 'Data analyst Senior Specialist ', 'Javascript developer',
 'Graphic Designer ', 'UI UX designer ',
 'Software testing team lead ', 'Junior front end developer ',
 'Django developer', 'Technical Engineer',
 'Fullstack Developer (laravel & flutter)', 'flutter developer',
 ',
 'Junior developer .net', 'It help desk', 'Physical Therapist ',
 'Senior interior designer ', 'tester',
 'Full stack web developer . Net & angular ', 'Business
 Analyst',
 'Full Stack .Net Developer', 'System administrator',
 'DevOps engineer ', 'Business Intelligence Specialist ',
 'Digital marketing specialist', 'Backed developer c sharp',
 'Front-end Developer', 'Web GIS developer', 'Back end laravel',
 ',
 'Frontend developer ', 'Senior sales ',
 'SENIOR CYBER THREAT ANALYST ', 'Php laravel developer',
 'Front End Developer React', 'Business analysis',
 'Odoo Developer ', 'Data annotation team leader ',
 'Business development manager ', 'Senior Product Marketing',
 'Senior Data Scientist', 'Mid level .net developer',
 'Junior Product owner', 'Cost accountant',
 'senior frontend engineer', '.net (desktop developer)',
 'Backend Engineer', 'Full stack javascript developer ', 'Typist',
 ',
 'Senior software quality control engineer', 'UI/UX Designer',
 'It help desk ', 'MEARN STACK -Front end Developer ',
 'Desktop app developer.Net', 'UI UX Designer',
 'Teamlead backend developer PHP', 'Customer service officer ',
 'Junior PHP Developer', 'AI engineer ', 'Angular developer ',
 'Supply chain analyst ', 'IT Security Engineer ',
 'Sales Data Analyst', 'Data analytics ',
 'Quality Assurance Senior Specialist', 'Warehouse Supervisor ',
 'سيلز', 'Marketing Director', 'Bi Developer ', 'Internal
 audit',
 'Senior Application Engineer', 'Service desk agent',
 'NLP Engineer', 'Clinical pharmacist ', 'Sales ',
 'Odoo implementer', 'Full Stack Web Developer',

```

'System Administrator', 'DBA ',
'Product specialist (medical sales)', 'Frontend',
'Full stack web developer ', 'Full stack developer python ',
'Sales Analyst', 'Odoo', 'React.js developer',
'Fullstack intern - Java & Angular',
'Senior Software Developer php', 'java developer',
'(AKA) DevOps Engineer', 'Django developer ', 'Sales
coordinator ',
'Siebel CRM developer ', 'Wordpress developer',
'Unity Game Developer ', 'laravel developer',
'Full Stack Node Angular', 'Data Engineer intern',
'Full stack .Net and react developer',
'Software Development team leader', 'Software support engineer
',
'Software QA', 'Embedded software engineer ',
'Senior Android-Flutter', 'Business developer ',
'Teaching assistant ', 'Senior dot net developer ',
'.NET developer', 'Front-End Developer ',
'Principal Dotnet Developer ',
'Associate SW Development Consultant',
'Full stack developer using .Net',
'Business Intelligence Developer', 'Junior softwrae Engineer',
'Sales account manager ', 'HR specialist ', 'Internal control
',
'Data analysis and it support',
"I'm sorry.(I'm saas) account manager.and.salaries", 'Beginner
',
'Quality control and automation engineer ', 'Product specialist
',
'Fluuter developer', 'Data analyst and operation executive ',
'Frontend Developer | React Developer ',
'director of preventive medicine ',
'Mobile developer Android and iOS ', 'Junior Data Scientist',
'Font End Developer (Angular)', 'Software developer ',
'Embedded SW Engineer ', 'Seniors interior design'],
dtype=object)

```

7. How many different elements are there?

```
df['Title'].nunique()
```

704

8.

midrange = (max-min)/2

range = The difference between max and min → shows the range of values

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 908 entries, 0 to 907
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Unnamed: 0                            908 non-null    int64
1   TimeStamp                             908 non-null    object
2   Title                                 908 non-null    object
3   Years of Experiences                  907 non-null    object
4   Salary                                908 non-null    object
5   Date of Salary                        908 non-null    object
6   What Is your Company                 908 non-null    object
7   Work Type                            908 non-null    object
8   Work Hour                            908 non-null    object
9   City of Company site                 816 non-null    object
dtypes: int64(1), object(9)
memory usage: 71.1+ KB

```

```

# 9. Numerical Attributes
# correlation

```

```

numericalarr = ['Salary' , 'Years of Experiences']

```

```

df[numericalarr].corr()

```

```

-----
-----
KeyError                                Traceback (most recent call
last)
Cell In[77], line 6
      1 # 9. Numerical Attributes
      2 # correlation
      4 numericalarr = ['Salary' , 'Years of Experiences']
----> 6 df[numericalarr].corr()

File c:\Python314\Lib\site-packages\pandas\core\frame.py:4119, in
DataFrame.__getitem__(self, key)
    4117     if is_iterator(key):
    4118         key = list(key)
-> 4119     indexer = self.columns._get_indexer_strict(key, "columns")
[1]
    4121 # take() does not accept boolean indexers
    4122 if getattr(indexer, "dtype", None) == bool:

File c:\Python314\Lib\site-packages\pandas\core\indexes\base.py:6212,
in Index._get_indexer_strict(self, key, axis_name)
    6209 else:
    6210     keyarr, indexer, new_indexer =
self._reindex_non_unique(keyarr)
-> 6212 self._raise_if_missing(keyarr, indexer, axis_name)

```



```

6214 keyarr = self.take(indexer)
6215 if isinstance(key, Index):
6216     # GH 42790 - Preserve name from an Index

```

```

File c:\Python314\Lib\site-packages\pandas\core\indexes\base.py:6264,
in Index._raise_if_missing(self, key, indexer, axis_name)
    6261     raise KeyError(f"None of [{key}] are in the
[{axis_name}]")
    6263 not_found = list(ensure_index(key)[missing_mask.nonzero()
[0]].unique())
-> 6264 raise KeyError(f"{not_found} not in index")

```

KeyError: "['Years of Experiences'] not in index"

Section 2

visualization

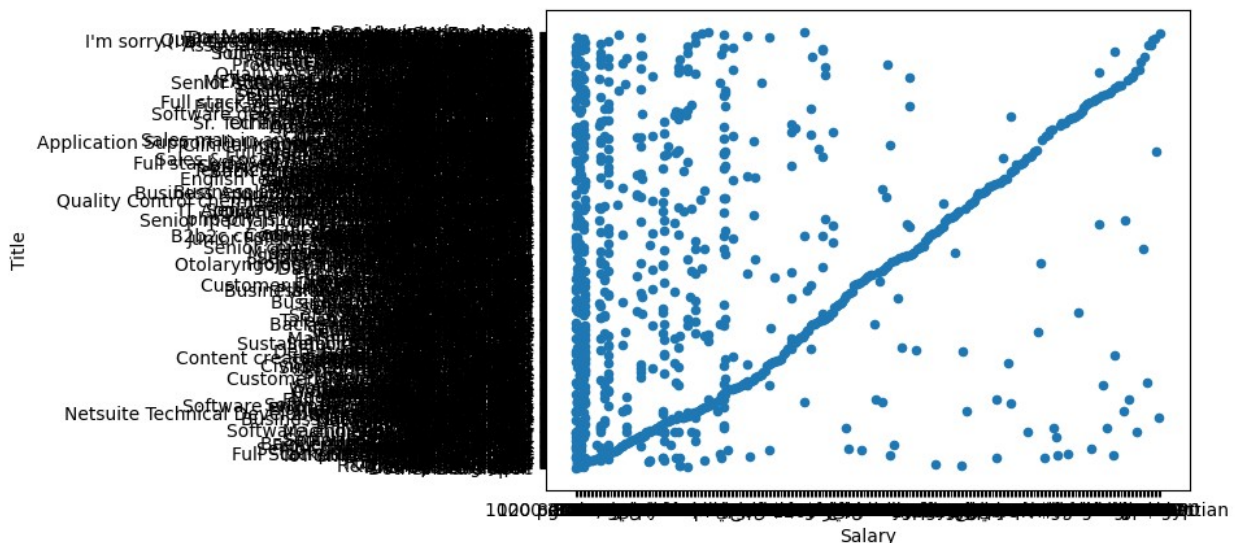
```

# 1. Data Visualization
import matplotlib.pyplot as plt

# 1.1 Scatter Plots
df.plot( x = 'Salary' , y = 'Title' , kind = 'scatter')

<Axes: xlabel='Salary', ylabel='Title'>

```



```

# 1.2 Line Plots
df.plot(x = 'Salary' , y = 'Title' , kind = 'line')

# columns must be numeric

```

```
# 1.3 Area Plots
df.plot(x = 'Salary' , y = 'Title' , kind = 'area')

# columns must be numeric

# 1.4 Bar Charts
df.plot( x = 'Salary' , y = 'Title' , kind = 'bar')

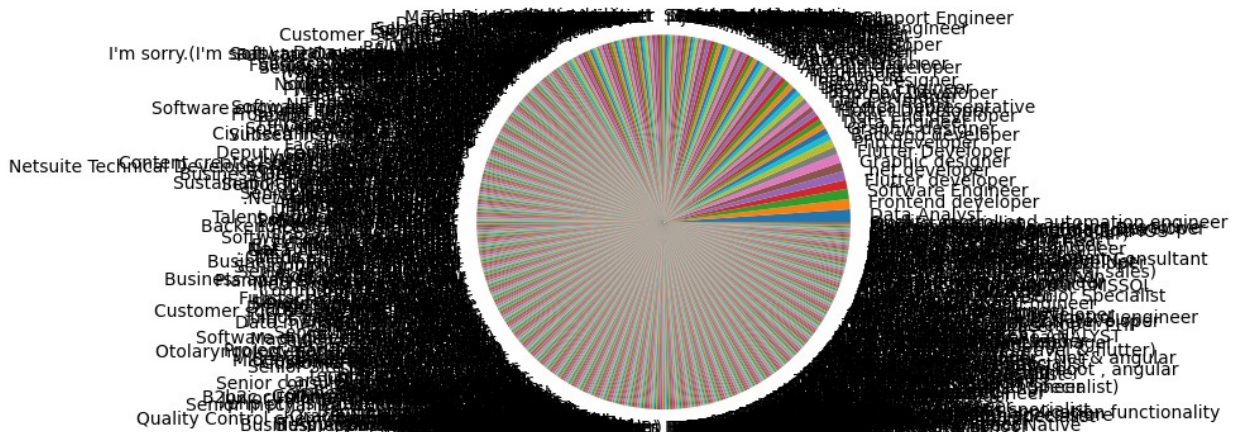
# kind = 'barh'
# # columns must be numeric

# 1.5 Histograms
df.plot( x = 'Salary' , y = 'Title' , kind = 'hist')

# 1.6 Box Plot
df['Salary'].plot(kind = 'box')

# 1.7. Pie Charts
df['Title'].value_counts().plot(kind = 'pie')

<Axes: ylabel='count'>
```



```
# 1.8 Color Map (Colored Scatter Plot)
df.plot.satter( x = ' ', y = ' ', c = ' ' )
```

Data Preprocessing

1. DEALING WITH MISSING VALUES

```
# How to discover missing values?
df.isnull()
```

	Unnamed: 0	TimeStamp	Title	Years of Experiences	Salary	\
0	False	False	False	False	False	
1	False	False	False	True	False	

2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
..	...	...	...	...	...	...
903	False	False	False	False	False	False
904	False	False	False	False	False	False
905	False	False	False	False	False	False
906	False	False	False	False	False	False
907	False	False	False	False	False	False

	Date of Salary	What Is your Company	Work Type	Work Hour \
0	False	False	False	False
1	False	False	False	False
2	False	False	False	False
3	False	False	False	False
4	False	False	False	False
..	...	...	...	...
903	False	False	False	False
904	False	False	False	False
905	False	False	False	False
906	False	False	False	False
907	False	False	False	False

	City of Company site
0	False
1	False
2	False
3	False
4	False
..	...
903	False
904	False
905	False
906	False
907	False

[908 rows x 10 columns]

```
df.isnull().sum()
```

Unnamed: 0	0
TimeStamp	0
Title	0
Years of Experiences	1
Salary	0
Date of Salary	0
What Is your Company	0
Work Type	0
Work Hour	0

```

City of Company site      92
dtype: int64

# Handling Missing Values
df2 = df.dropna()

df2.isnull().sum()

Unnamed: 0                0
TimeStamp                 0
Title                    0
Years of Experiences      0
Salary                   0
Date of Salary            0
What Is your Company      0
Work Type                 0
Work Hour                 0
City of Company site      0
dtype: int64

# Keep the row if it has at least 3 correct values.

df2 = df.isnull().dropna(thresh=3)

# Filling with Constant
df3 = df2.fillna('NA')

```

important

```

# missing values في عمود معين من data frame :
# loc بنستخدم

import numpy as np
df.loc[0:4 , 'Salary'] = np.nan

# dataframe في عمود معين من outliers لو طلبت نضيف :
# loc بنستخدم برده

df.loc[13 , 'Salary'] = 100000
df.loc[15 , 'Salary'] = 500000
df.loc[14 , 'Salary'] = 900000

```

2. DEALING WITH OUTLIERS

```

# Detect Outliers Using IQR
df2['Salary'].plot(kind = 'box')

Q1 = df['Salary'].quantile(0.25)
Q3 = df['Salary'].quantile(0.75)

```

```
IQR = Q3 - Q1
print(IQR)

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

df_clean = df[(df['Salary'] >= lower_bound) & (df['Salary'] <=
upper_bound)]
```

Dimension Elimination

You have four different ways to perform Dimension Elimination:

1. Slicing columns.
2. Drop columns by position.
3. Selecting important columns.
4. Deleting unimportant columns by name.

```
# Data Sampling

sample1 = df['Salary'].sample(n = 5)

# percentage
sample2 = df['Salary'].sample(frac = 0.1)
```

Data Scaling

min-max normalization

z-score

NEAREST NEIGHBOR CLASSIFIERS

KNN is a Supervised Machine Learning algorithm used for Classification and Regression.

- It relies on the principle that data with similar characteristics often belong to the same class.
- How it Works? When a new point is reached:
 1. We calculate the distance between it and all points in the data.
 2. We select the nearest K points.
 3. The most frequently occurring class among them is the result

Radius Neighbors (RNN)

- Radius Neighbors is an extension of the KNN algorithm. However, instead of choosing a fixed number of

neighbors (K), we select all points within a specified distance (Radius) from the new point.

- How it Works?

1. We determine the value of the radius.
2. We search for all points within this distance.
3. We use them to determine the classification

```
# Iris Dataset.
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets

iris = datasets.load_iris()
```

data splitting

```
from sklearn.model_selection import train_test_split

X = df.drop('Salary', axis=1)
y = df['Salary']

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

# 1. استدعاء المكتبات الأساسية
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix

# 2. 'target' بافتراض إن اسم عمود التارجت هو) تجهيز البيانات
X = df.drop('target', axis=1)
y = df['target']

# 3. تقسيم الداتا
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

# 4. استدعاء الموديل (تغيري السطر ده حسب الموديل المطلوب)
# لو Decision Tree:
from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier()

# لو Naive Bayes:
# from sklearn.naive_bayes import GaussianNB
# model = GaussianNB()
```

```
# 5. التدريب (الخطوة الثابتة لأي موديل)  
model.fit(X_train, y_train)
```

```
# 6. التوقع والتقييم  
y_pred = model.predict(X_test)  
print(f"Accuracy: {accuracy_score(y)}
```

```
### لو طلب test, y_pred}))"  
print(f"Confusion Matrix: \n{confusion_matrix(y_test, y_pred)}")
```